

KEY



MATH 241-Spring 2014

Name : _____

QUIZ 2

Student ID : _____

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Section : _____

1. Let V be a vector space over the real numbers. Assume $\{u, v, w\}$ is a linearly independent subset of V . Determine whether the following subsets of V are linearly independent, give your reasoning.

(a) $\{u+v, v+w, w+u\}$

(b) $\{u-v, v-w, w-u\}$

a) let $\alpha(u+v) + \beta(v+w) + \gamma(w+u) = 0$ where

$\alpha, \beta, \gamma \in \mathbb{R}$.

Then, $(\alpha + \gamma)u + (\alpha + \beta)v + (\beta + \gamma)w = 0$

As $\{u, v, w\}$ are linearly independent,

$$\begin{cases} \alpha + \gamma = 0 \\ \alpha + \beta = 0 \\ \beta + \gamma = 0 \end{cases} \Rightarrow \text{then } \alpha = \beta = \gamma = 0 \text{ (either by solving simultaneously or by noting the system, both are okay)}$$

is the unique soln.

$\Rightarrow \{u+v, v+w, w+u\}$ are lin. indep.

b) let $\alpha(u-v) + \beta(v-w) + \gamma(w-u) = 0$, $\alpha, \beta, \gamma \in \mathbb{R}$

Then, $(\alpha - \gamma)u + (-\alpha + \beta)v + (-\beta + \gamma)w = 0$

As $\{u, v, w\}$ are lin. indep.,

$$\begin{cases} \alpha - \gamma = 0 \\ -\alpha + \beta = 0 \\ -\beta + \gamma = 0 \end{cases} \Rightarrow \begin{cases} \alpha = t \\ \gamma = t \\ \beta = t \end{cases} \text{ is a soln for any } t \in \mathbb{R}.$$

In particular for $\alpha = \beta = \gamma = 1$, $1 \cdot (u-v) + 1 \cdot (v-w) + 1 \cdot (w-u) = 0$
 $\Rightarrow \{u-v, v-w, w-u\}$ are linearly dependent